Keeper tier-gate — three Saturday afternoon items: Lyra bulk-color v0.4 route commitment (C+D), Elie engine v0.2 audit support v0.2 (15 toys 3612-3626), Lyra T1 falsifier v0.1 (lepton count). Bulk-color: COMMITMENT ACCEPTED at FRAMEWORK tier with two conditions (Toeplitz-source sourcing + AdS/CFT analogy demotion). Engine support: PASS; §6/§7 candidates STRUCTURAL pending engine v0.3 consolidation doc; |V_cb| flagged for Cal cold-read with Cal #27 brake on denominator. T1: PASS as falsifier design with one trigger-sharpening note for v0.2. None promoted to RATIFIED; all three remain in active queue.

Keeper

2026-05-30 Saturday

Tier-gate — bulk-color v0.4 + engine support v0.2 + T1 falsifier v0.1

0. Scope

Three items pulled from the Saturday afternoon queue: 1. Lyra bulk-color v0.4 route commitment (C+D) — A1 macro frontier 2. Elie engine v0.2 audit support v0.2 — 15 toys 3612-3626 consolidated 3. Lyra T1 falsifier v0.1 — lepton-sector count from canonical-basis dim

All three reviewed end-to-end. Verdicts and conditions below.

1. Bulk-color v0.4 — Route Commitment (C+D)

Verdict: COMMITMENT ACCEPTED at FRAMEWORK tier. The (C+D) unification is structurally well-motivated; multi-week derivation burden honest; framing passes Cal #27 (Lyra explicitly says "I am NOT claiming SU(3) emergence is derived").

Strengths: - (D) = SO(3) ⊂ SO(5) sub-vector for 3-fold counting (from v0.2, already tier-gated CANDIDATE). - (C) = Toeplitz-algebra on $H^2(S)$ as source of non-abelian structure compatible with [p_+, p_+]=0 (Elie A1 obstruction). Real mathematical structure, not speculation. - (C+D) complementarity argued cleanly: (D) provides count labels, (C) provides gauge structure acting on labels. - Multi-week deferral list explicit (Toeplitz computation; SU(3) identification; 8-gluon adjoint; QCD low-energy emergence).

Conditions (must be addressed before any external use):

- C1 — Toeplitz-source sourcing: "Toeplitz algebras on Hardy spaces of bounded symmetric domains are well-studied (Upmeier, Berezin, Vasilevski)" is currently RECALLED, not VERIFIED-CITED (Cal #33

STANDING). For the load-bearing claim "T(H^2(S)) is generally non-commutative even when underlying boundary multiplication is abelian," v0.5+ must cite the specific theorem (preferred: Upmeier, *Toeplitz Operators and Index Theory in Several Complex Variables*, or Vasilevski's volumes — with page/theorem number) demonstrating this for the Shilov boundary of D_IV^5 specifically, OR for a bounded symmetric domain of Type IV with adaptation argument.

- C2 — AdS/CFT analogy demotion: "Substantively similar to the AdS/CFT-style boundary algebra giving bulk dynamics" should be demoted in v0.5+ to "evocative parallel, not structural argument." AdS/CFT operates on AdS, not on D_IV^5; the analogy is suggestive but cannot do structural work here. Currently the doc treats it as parallel-of-the-mechanism; this needs to be marked as motivational only.

Tier disposition: - Route (C+D) commitment: FRAMEWORK - Underlying math (Toeplitz/Hardy on bounded symmetric domains): RECALLED, pending verification per C1 - SO(3) sub-vector counting (D): CANDIDATE (carried from v0.2) - SU(3) gauge emergence from (C+D): OPEN, multi-week

Status: A1 macro frontier ACTIVE; (C+D) is the favored direction; Lyra/Elie/Cal triangle continues the work.

2. Engine v0.2 audit support v0.2 — 15 toys 3612-3626

Verdict: PASS. Honest tier-disposition table throughout; cross-toy substrate-primary coherence check is strong evidence of structural distribution, not concentration. Engine v0.2 stands at PASS (K1-K3 conditions from prior Keeper audit substantively cleared by toys 3609 + 3610); engine v0.3 §6/§7 candidates ready when Elie files the consolidation doc.

Substantive findings absorbed:

- Engine v0.3 §6 candidate (Drinfeld double + CPT, Toy 3617): F-mirror exact; Drinfeld pairing carries N_c, N_c·n_C; $\omega/\sigma/W_0$ map to C/T/P. STRUCTURAL acceptance pending engine v0.3 consolidation. K1·2 audit anchor reserved.
- Engine v0.3 §7 candidate (bulk-color algebraic, Toy 3620): SO(5) ⊃ SO(3)×SO(2) decomposes 5 = N_c + rank; SM gauge h^∨ counts match (SU(3):3=N_c, SU(2):2=rank). STRUCTURAL. K1·4 audit anchor reserved. Cross-confirms Lyra bulk-color v0.2/v0.4 SO(3) sub-vector reading.
- |V_cb| ↔ T2442 cross-anchor (Toy 3622): candidate substrate form 225/5480 = $(N\_c{\cdot}n\_C)^2$/(2^N_c·n_C·N_max), 0.04σ from observed. Numerator clean (225 cross-anchored in 6 domains). Cal #27 BRAKE on denominator: 2^N_c·n_C·N_max is less structurally forced than the numerator; the 0.04σ precision should not headline. Disposition: CANDIDATE with cross-anchor noted; flag for Cal cold-read; load-bearing claims capped at MATCHED.
- PMNS 3/3 substrate fractions (Toy 3618): all three angles within 1σ as n/N_max with substrate-natural numerators (rank·N_c·g=42; $N\_c{\cdot}n\_C^2$=75; N_c=3). VERIFIED at PDG; F1 falsifier LIVE; JUNO+DUNE tightening expected.
- Bell sub-Tsirelson 1/2^N_c = 1/8 = 12.5% (Toy 3626 T2469 SCMP): PREDICTION at SCMP T2469; falsifiable at current Bell precision ~1% on S. Strong falsifier; SP-30 Vienna response pending per memory.
- 6-domain 225 fingerprint extended (Toy 3622 + catalog): Bergman + Ag Debye + Cu/Ag + Ag/Pb + cosmological c_reg + CKM |V_cb|. Cross-scale substrate constant. STRUCTURAL.
- 18-cell substrate spine in 66-K-type table (Toy 3614): backbone for Grace Periodic Table v0.5+ work. STRUCTURAL backbone.
- B_2-specificity of E7 mult-3 (Toy 3615, A_2 comparison): A_2 gives mult-0; B_2 gives mult-3 with substrate-primary Casimirs (0, rank^2, C_2). Strengthens generation count derivation from h^∨=N_c=3. Honest qualifier: quantitative m_μ/m_e NOT derived — Lyra L4 v0.2 lane.

Sourcing discipline check (Cal #33): Elie stayed in command (finite-B_2 + SO(5) Clebsch-Gordan + q-Serre + Cartan HSD classification); declined affine species + Bergman-kernel-explicit derivations. Passes.

Honest scope items noted: timestamp self-correction at 09:35 EDT; Cal #27 applied with CD caveats; no external promotion; sustainable cadence (15 toys morning, standing reactive afternoon). All pass.

Status: Engine v0.2 PASS holds; engine v0.3 §6+§7 candidates STRUCTURAL; K1·2 + K1·4 audit anchors pre-reserved for when Elie files the consolidation doc.

3. T1 falsifier v0.1 — lepton-sector count

Verdict: PASS as falsifier design with one trigger-sharpening note for v0.2.

Strengths: - Pure-QG layer (dim V_(1/2,1/2) of $U\_q(B_2)$ at q=2 = 4) is rigorous Weyl formula, no dictionary leakage. Currently matches observed Dirac 4-spinor structure ?. - Dictionary-combined layer ($24 = 4 \times 2$ doublet $\times$ 3 generations) is explicitly layered per Cal #27. Matches observed Dirac SM with 24 lepton components ?. - Falsifier triggers explicit YES/NO across five channels (4th gen, $0\nu\beta\beta \to$Majorana, non-doublet, SUSY partners, extra-K-type leptons). - F5 (no SUSY, no 4th gen, Dirac neutrinos) cross-links cleanly. - The two-layer framing (pure-QG vs dictionary-combined) per Cal #27 is exactly right — the cleanest pure-QG channel is dim 4 = Dirac, which is currently confirmed.

Note for v0.2 sharpening (Cal cold-read recommended):

- 4th-generation trigger refinement: "4th generation discovered → 32 components → CONTRADICTS" assumes the hypothetical 4th gen comes as a doublet of Dirac fermions. If instead it appeared as right-handed sterile only (e.g., 2 components per particle), the count would not be 32. The trigger should specify "4th generation of doublet-organized Dirac leptons" to be unambiguous.
- Extra-K-type trigger refinement: "K-type beyond V_(1/2,1/2) hosts a charged lepton" needs to specify what would constitute an observable signature — the experimental statement should reference a measurable consequence (e.g., a heavy lepton with anomalous spin structure, distinct from a 4th generation).
- $0\nu\beta\beta$ trigger: clean as stated; LEGEND-1000 and nEXO are the experiments.

These are minor; the falsifier design v0.1 is substantively sound and is one of the cleaner quantitative tests of the keystone bet.

Tier disposition: - Pure-QG dim 4 falsifier: PASS, MATCHED at Dirac 4-spinor structure - Dictionary-combined 24-count falsifier: PASS, MATCHED at SM Dirac scenario; F5 cross-link active - Triggers: 5 channels LIVE; recommend v0.2 sharpening per the two notes above

Status: T1 falsifier joins F1 (PMNS) + Bell sub-Tsirelson 1/8 (T2469 SCMP) as the three sharpest quantitative falsifiers currently in flight. All three carry through to v0.2 sharpening + Cal cold-read.

4. Queue advance + handback

Three items cleared from my queue. Status:

- A1 paper v0.4 Final Disposition is PASS per Lyra 09:21 doc + CLAUDE.md headline ("A1 Substrate Hall Algebra (PRIMARY) = Keeper final PASS v0.4"); my queue-#1 "A1 v0.5" was a phantom — that work is closed pending Cal quick-verify + proof-stage genus citation.
- Bulk-color v0.4 commitment ACCEPTED at FRAMEWORK with C1+C2 conditions for v0.5+.
- Engine support v0.2 PASS with §6+§7 STRUCTURAL candidates pre-reserved.
- T1 falsifier v0.1 PASS with v0.2 sharpening notes routed to Cal.

Pulling next from my queue: - Cold-read on Two-Route Overdetermination Scan v0.1 (Grace, 10:00) — Saturday morning batch highest-leverage novel result - Cold-read on quasi-eigentone framework v0.1 (Lyra, 09:58) - Cold-read on Vol 16 Ch1 §A_sub-Hall Dictionary v0.1 (Lyra, 10:01) - Engine v0.2 → v0.3 RE-AUDIT when Elie files the consolidation doc itself (the audit-support doc here is preparatory; the engine v0.3 doc with §6+§7 sections is the audit target) - K-audit cascade through the queue: spectral score theoretical layer, Vol 16 §3, L7 baryogenesis, L8 α-placement, L9 Macdonald-Koornwinder, Two-Route Scan formalization

Routed: - Lyra: bulk-color v0.5 absorbs C1+C2 (Toeplitz-source citation; AdS/CFT demotion); T1 v0.2 sharpens 4th-gen + extra-K-type triggers per Cal cold-read. - Elie: engine v0.3 consolidation doc with §6 (Drinfeld/CPT, from Toy 3617) + §7 (bulk-color algebraic, from Toy 3620) when ready for K1·2 + K1·4 audit. - Cal: cold-read queue extends — |V_cb| candidate denominator (Cal #27 brake applied; verify), T1 falsifier triggers (4th gen + extra K-type), bulk-color C+D commitment (Toeplitz-algebra soundness for D_IV^5 specifically). - Grace: 18-cell substrate spine absorbed into Periodic Table v0.5+; 6-domain 225 fingerprint maintained as cross-anchor evidence.

— Keeper. Three tier-gates filed Saturday afternoon; queue advanced; conditions clean; A1 macro frontier (bulk-color) carries forward to multi-week derivation with (C+D) committed.